

JAMES CHRISTOPHER HARTWELL

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PROFESSIONAL SUMMARY

Results-driven and highly motivated Principal Engineer with 8 years of proven track record spearheading cutting-edge technology initiatives and delivering actionable insights through data-driven decision making. Passionate about synergistically leveraging innovative solutions to champion mission-critical organisational objectives and cross-functional collaboration. Demonstrated expertise in architecting scalable systems and fostering dynamic technical excellence to drive organisational transformation through strategically implemented solutions.

WORK EXPERIENCE

Principal Engineer

TechVenture Solutions Ltd

<https://www.techventuresolutions.io>

Manchester, United Kingdom

January 2022 - Present

Architected and deployed microservices infrastructure supporting 40+ internal teams, reducing system latency by 35% and improving deployment frequency to 15+ releases per day across the organisation.

Led the implementation of machine learning pipelines using HuggingFace Transformers and dbt for real-time predictive analytics, enabling data-driven decision making across finance and operations departments.

Established engineering standards and best practices for cloud-native development, mentoring 12 junior engineers and ensuring adherence to security frameworks including OWASP Top 10.

Championed the migration from monolithic architecture to event-driven systems using Apache Kafka, reducing operational costs by 28% and improving system resilience to 99.99% uptime.

Senior Software Architect

DataFlow Systems Inc

<https://www.dataflowsystems.net>

Manchester, United Kingdom

November 2020 - March 2021

Designed and implemented real-time ETL pipelines processing 500GB+ daily data volumes, reducing query response times from 45 seconds to under 3 seconds for executive dashboards.

Established CI/CD infrastructure using GitHub Actions and GitLab pipelines, automating deployment processes and reducing release cycles from monthly to weekly deployments.

Led cross-functional teams of 8 engineers to architect data warehouse solutions on AWS, delivering 99.95% availability and supporting 200+ concurrent users.

Managed database optimisation initiatives across PostgreSQL and MongoDB clusters, achieving 40% improvement in query performance and reducing infrastructure costs by 22%.

Lead Software Engineer

Innovate Digital Group

<https://www.innovatedigitalgroup.co>

Manchester, United Kingdom

June 2016 - October 2020

Developed and maintained microservices architecture supporting 2 million daily active users, using Python, Go, and Node.js across containerised Docker environments.

Implemented automated testing frameworks reducing production incidents by 67% and improving code coverage from 52% to 89% across the platform.

Mentored team of 6 engineers through technical design reviews and code audits, establishing coding standards and architectural patterns.

Optimised payment processing systems handling GBP 50+ million in monthly transactions, improving transaction success rates to 99.97% and reducing fraud detection false positives by 45%.

EDUCATION

BSc Computer Science (First Class Honours)

University of Manchester

Manchester, United Kingdom

2012 - 2015

TECHNICAL SKILLS

Programming Languages: Python, Go, Java, TypeScript, SQL

Cloud Platforms: AWS (EC2, S3, Lambda, RDS), Google Cloud Platform, Azure

Data Technologies: PostgreSQL, MongoDB, Apache Spark, Apache Kafka, Elasticsearch

DevOps and Infrastructure: Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, GitLab CI/CD

Machine Learning: TensorFlow, HuggingFace Transformers, Scikit-learn, dbt

Architectural Patterns: Microservices, Event-Driven Architecture, CQRS, Domain-Driven Design

Security and Compliance: OAuth 2.0, JWT, OWASP, SOC 2, ISO 27001